

This version of your documentation is optimized for **Text-to-Speech (TTS)** software. I have converted all abbreviations, technical units, and mathematical notations into full words to ensure the AI voice reads them naturally and accurately. I have also removed all Em dashes as requested.

SAGE Hardware Documentation Questionnaire

For: Mechanical and Electrical Engineers

This document contains the questions you need to answer to produce complete hardware documentation for SAGE. The goal is that someone with no prior exposure to the robot, a new student, a faculty member, or an artificial intelligence assistant, can read the resulting document and understand exactly how SAGE is physically built, how its electronics are connected, and how to diagnose or repair any hardware problem.

Part 1: Chassis and Mechanical Structure

1.1 Frame Overview

The structural integrity of SAGE relies on a hybrid design combining a rigid aluminum skeleton with high-strength three D printed components.

Dimensions and Mass

- Length: 615 millimeters
- Width: 629 millimeters
- Height: 625 millimeters

3D Printing Specifications

Most components were produced on a Bambu Lab A1 using the zero point two zero millimeter Strength profile.

- Material: PETG, chosen for its durability and heat resistance.
- Wall Loops: 6 loops for increased structural rigidity.
- Infill: 25 percent infill.

Structural Components

The robot uses aluminum extrusions as the primary internal frame to prevent the chassis from flexing under the weight of the battery and motors.

Major Assemblies

There are eight distinct three D printed assemblies that make up the robot:

- Base Platform: The foundation of the robot.
- Middle Frame: The central structural shell.

- Lid: Houses the primary sensors, including the LiDAR and Camera.
- Wheel Brackets: Mounts for the front omni wheels.
- Motor Assembly: Houses the VersaPlanetary drive motors and Nidec encoders.
- Speaker Mount: Secures the audio system.
- Battery Holder: A reinforced frame for the 56 cell pack.
- Jetson Holder: A snug-fit mount for the primary computer.

Fasteners and Mounting

The robot is held together using a mix of imperial and metric hardware:

- Standard Structural Joints: One fourth dash twenty screws paired with nylon lock nuts for high-load areas.
- General Assembly: Eight dash thirty two screws for smaller printed components.
- Aluminum Integration: Metric T-nuts and M4 screws are used to secure printed parts to the aluminum extrusion channels.
- Jetson Orin Nano: Mounted using a friction-fit snug three D printed holder. This design allows for quick removal without tools while keeping the board secure during movement.
- STM32 Board: The microcontroller is mounted directly to the custom printed circuit board breakout board via standard two point five four millimeter header pins, allowing for easy replacement or debugging if necessary.

1.2 Wheels and Drive System

The robot utilizes a differential drive configuration with a combination of traction wheels for power and omni wheels for maneuverability and precise tracking.

Drive and Support Wheels

The main movement of the robot is handled by high-traction rear wheels, while the front remains agile using passive rollers.

- **Driven Wheels (Rear):**
 - Model: 6 inch MAXSpline Hard Grip Wheel
 - Part Number: REV-21-2446
 - Details: 6 inch diameter, hard-tread material designed for consistent grip on flat surfaces.
- **Passive Wheels (Front):**
 - Model: 6 inch MAXSpline Omni Wheel
 - Part Number: REV-21-2476
 - Details: 6 inch diameter, allowing the front of the robot to slide laterally during turns for a zero-turn radius.

Dead Wheels (Odometry)

For high-precision position tracking, the robot uses a dedicated dead wheel system that is independent of the drive motors. This prevents wheel slip from affecting the robot's calculated location.

- Components: Two 4 inch omni wheels.
- Mounting: The wheels are mounted on pivoting arms attached to the aluminum frame.
- Ground Contact: To ensure the wheels never lose contact with the floor, which would cause tracking errors, a rubber band is tensioned on one end of the pivot. This creates a spring-loaded mechanism that constantly pulls the wheel against the ground.

Drive Geometry

The distance between the wheels is a critical value for the robot's turning math.

- Wheel Track: The center-to-center distance between the two driven wheels is zero point three eight one meters. This value is used by the firmware to calculate the robot's heading and rotation.

Part 2: Electrical System and Power

2.1 Battery

Battery Specifications

The robot is powered by a custom lithium-ion battery pack. It uses 56 individual 18650 cells, specifically the Samsung 35E, arranged in a four S fourteen P configuration.

- Part Number: Samsung INR18650-35E
- Total Capacity: 49 Amp hours
- Total Energy: 725 Watt hours
- Voltage Range: 12 Volts to 16.8 Volts
- Maximum Discharge Rate: 2.29 C

Runtime and Maintenance

- Expected Runtime: Approximately 12 hours under normal operating conditions.
- Minimum Safe Voltage: The battery should be recharged before it drops below 12 Volts, which is 3.0 Volts per cell, to ensure long-term health, though the Battery Management System provides a hard cutoff at 10.2 Volts.

Physical Installation and Connection

The battery is housed in a three D printed holder and is securely mounted to the robot's aluminum frame. To distribute power:

1. The cells connect through a Battery Management System to a main fuse.
2. A terminal block splits the power to the Jetson computer, motors, and other electronics.
3. An ideal diode is used to manage the connection to the external charging contacts safely.

Battery Management System Specifications

- Maximum Continuous Current: 30 Amps
- Over-Current Protection: 60 Amps
- Over-Charge Cutoff: 4.28 Volts per cell
- Over-Discharge Cutoff: 2.55 Volts per cell
- Operating Temperature: negative 30 degrees Celsius to plus 80 degrees Celsius

Charging Procedure

- Method: Contact-based charging.
- Maximum Charging Current: 10 Amps.
- Procedure: The robot aligns its charging contacts with the dock. The internal ideal diode prevents back-feeding power while charging is in progress.

2.2 Power Distribution

The power system originates from the four S lithium-ion battery pack and is distributed through a central terminal block. A 30 Amp fuse is placed between the Battery Management System and the terminal block for circuit protection.

Power Rails and Consumers

The system utilizes a direct battery rail for high-power components and regulated rails for sensitive electronics:

- **Main Battery Rail (12 Volts to 16.8 Volts):**
 - Jetson Orin Nano: Powered directly via the barrel jack.
 - Motor Electronic Speed Controllers: Connect directly to provide power to the drive motors.
 - DC to DC Buck Converter for LEDs: Steps down battery voltage for the lighting system.
 - DC to DC Buck Converter for iPad: Provides regulated power for iPad charging.
- **Regulated Low Voltage Rail:**
 - Linear Voltage Regulator: Provides stable power to the STM32 board and encoders.
- **Regulated 5 Volt Rail via Jetson USB:**
 - RPLIDAR: Powered through the Jetson's USB port.
 - Speaker: Powered and audio-driven through the Jetson's USB port.

2.3 Controls and Protection

- Main Power Switch: Rocker switch located under the left side of the chassis.
- Overcurrent Protection: A 30 Amp main fuse and a secondary 60 Amp limit via the Battery Management System.

2.4 Motor System

Motor Specifications

The robot uses two VersaPlanetary motors, part number PM15R-25F-1001, for the drive system.

Electronic Speed Controllers

Both the left and right motors are controlled by identical speed controller units. These do not require a manual arming sequence and are functional immediately upon receiving a valid signal via standard pulse width modulation signal wires.

Pulse Width Modulation Configuration

- Forward Maximum: 1700 microseconds
- Stop: 1500 microseconds
- Reverse Maximum: 1300 microseconds
- These limits are hard-coded in the microcontroller firmware to ensure safe speed parameters.

Part 3: Electronics and Printed Circuit Board

3.1 STM32 Board

The robot uses an STM32G0B1 microcontroller. It is mounted on a custom printed circuit board designed by Aidan Matson and manufactured through JLC PCB.

3.2 Wiring

Wires are primarily managed using cable ties to keep them organized. All cables are routed and secured to the frame to ensure they do not interfere with moving parts. There are currently no known pinch points.

3.3 Encoder Wiring

Dead Wheel Encoders

- Model: Yosoo Health Gear Incremental Optical Rotary Encoder
- Resolution: 600 pulses per revolution
- Voltage: Operating at 5 Volts
- Output: AB 2-Phase Quadrature

Motor Encoders

The drive motors use 2-phase optical encoders to provide feedback for the PID loops.

Hardware Mapping

- Straight Deadwheel: Timer 1
- Horizontal Deadwheel: Timer 2
- Left Motor Encoder: Timer 3
- Right Motor Encoder: Timer 4

3.4 Battery Voltage Measurement

The system uses a resistor divider to step down the battery voltage of up to 16.8 Volts to a level safe for the microcontroller's Analog-to-Digital Converter.

- Resistor Values: 100 kilohms and 22 kilohms.
- Sensing Pin: PA4.
- Correction Factor: zero point nine eight four eight.

Part 4: Assembly and Maintenance

4.1 Assembly Order

Total assembly time is approximately 3 to 4 hours.

1. **Phase 1:** Assemble the three D printed base, aluminum frame, drive components, and odometry units.
2. **Phase 2:** Install the STM32 board, internal wiring, battery holder, and Jetson mount.
3. **Phase 3:** Assemble the middle shell and integrate the tablet.
4. **Phase 4:** Mount the LiDAR, camera, and microphone on the lid. Apply sound-deadening foam to the microphone.
5. **Phase 5:** Join the sections and secure the lid.

4.2 Critical Fastening Notes

- Thread-Locker: Blue Loctite is applied to high-vibration areas like drive wheels and deadwheel assemblies.
- Nylon Lock Nuts: Used for all other structural mounting points.
- Hot Glue: Used as secondary retention for electronic connectors like the motor encoder plugs.

4.3 Removing the Jetson

The Jetson Orin Nano is held by a friction fit. To remove:

1. Power down the main switch.
2. Disconnect USB cables and the barrel jack power.
3. Gently lift the unit vertically.
4. Static Safety: Touch a grounded part of the aluminum frame before handling the electronics.

Part 5: Safety and Handling

5.1 Electrical Safety

Wait at least one minute after powering off for the motor speed controller capacitors to discharge before touching electrical contacts. Never manually short the battery terminals.

5.2 Battery Handling

Store the battery at 50 percent state of charge if inactive for more than two weeks. Do not charge below 0 degrees Celsius, or 32 degrees Fahrenheit.

5.3 Operational Hazards

- Pinch Points: Keep fingers away from drive wheels when motors are armed.
- Speed Limits: Software-limited to one point five meters per second.
- Crush Hazard: SAGE is heavy and high-torque; it can easily crush small objects or cause tripping.